Ao (Allen) Xu

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Summary

Business data scientist with experience applying ML, NLP, and LLM workflows to support operations, seller analytics, case routing, and forecasting problems, translating ambiguous business needs into measurable data science solutions. Focused on AI-powered operations and decision systems, with a support-ticket routing case study evaluating automation coverage, routing confidence, human review load, and LLM cost tradeoffs.

Education

New York University — New York, NY
M.S. in Data Science | GPA: 3.5/4.0 | Teaching Assistant: Linear Algebra (2 yrs) — *Sep 2023 – May 2025*

University of California, Santa Barbara — Santa Barbara, CA
B.S. in Data Science & B.S. in Applied Mathematics | GPA: 3.81/4.0 — *Sep 2019 – Mar 2023*

Technical Skills

Programming & Data: Python (Pandas, NumPy, PySpark, Scikit-Learn), SQL, R
ML / Statistics: Logistic Regression, XGBoost, LightGBM, NLP, Feature Engineering, A/B Testing, Hypothesis Testing, Time-Series Forecasting, Model Evaluation
AI / LLM: LLM Evaluation, Prompt Engineering, RAG Workflows, OpenAI API, Dify, Google AI Studio, Claude Code, Cursor, Vector Databases (FAISS/Milvus)
Visualization / BI: Tableau, Power BI, Looker, Streamlit, Retool

Professional Experience

AI Product & Data Strategy Analyst, Project-based — Aug 2025 – Present
China Telecom AI Center | Remote / Shanghai

- Designed 3 LLM-powered workflows on Dify for intent classification, briefing generation, and narration; translated city-operations requirements into structured prompts, grounding rules, and measurable evaluation criteria.
- Built NLP pipeline classifying 20,000+ unstructured dialogue logs into 8 categories / 170+ intent labels, creating structured datasets for LLM evaluation, product analytics, and downstream workflow design.
- Designed prompt A/B tests and a data-quality framework combining manual labeling, LLM annotation, and stratified QA; achieved 90%+ label accuracy and generated 100K+ validation records for model and workflow testing.
- Prototyped a digital-human management platform by expanding a standalone API configuration page into resource, voice, capability, and workflow management modules; translated fragmented requirements into structured specs for engineering review and acceptance testing.

Data Analyst Intern — Jun 2024 – Aug 2024
TikTok (ByteDance) – Global Seller Operations — *| Beijing*

- Analyzed U.S. seller support and satisfaction data (SQL, Python, Tableau); identified a 10% MoM decline in onboarding satisfaction and traced support-channel KPI distortion to duplicate-ticket logic.
- Defined support KPIs including SSAT, 72-Hour Resolution Rate, ART, and Escalation Rate; mapped the end-to-end onboarding support workflow and identified 2 bottlenecks driving 84% of onboarding failures.
- Built Tableau dashboards and Python diagnostic scripts to monitor support-channel performance, surface data-quality issues, and give PM and operations teams a consistent view of seller onboarding health.
- Partnered with PM and sales-ops teams to operationalize fixes, improving resolution time by 51% and seller satisfaction score by 23%; quantified a 48% potential conversion-rate lift from funnel improvements.

Business Operations / Risk Advisory Intern — Jun 2021 – Aug 2021
Deloitte — *| Beijing*

- Built automated data workflow consolidating 3,500+ contract records, improving retrieval efficiency by 70%; developed SQL-based reporting logic and Power BI dashboards for portfolio-level risk operations monitoring.
- Standardized RFP execution workflows from requirements intake to QA, improving consistency of client-facing risk advisory deliverables.

Selected Projects

LLM-powered Support Ticket Routing System — GitHub | Live Case Study
Python, Scikit-Learn, NLP, Streamlit, LLM Evaluation, Kaggle

- Built a decision-oriented support-ticket routing workflow combining deterministic rules, calibrated ML, selective LLM fallback, and human triage to make automation coverage, routing confidence, human review load, and estimated LLM cost measurable.
- Developed supervised NLP benchmark on 8,325 deduplicated Kaggle support-ticket records with 1,665 held-out metadata-derived labels; TF-IDF / Logistic Regression outperformed keyword baseline across accuracy, macro-F1, and weighted-F1.
- Designed Streamlit case-study dashboard with routing simulator, cost–coverage threshold sweep, model evaluation, stakeholder view, and executive memo, translating ML tradeoffs into support-operations recommendations.

S&P Global × NYU Multi-source Financial Forecasting — Sep 2024 – Dec 2024
Python, SQL, PyTorch, LightGBM, XGBoost, Time-Series

- Designed multi-source forecasting pipeline integrating social-media sentiment with market indices across 62,800+ hourly observations to model financial market dynamics.
- Benchmarked 6 forecasting architectures, including ARIMA, SE-GRN, iTransformer, SOFTS, CNN-LSTM, and Times-FM; delivered champion SOFTS model with 71% MAE reduction over baseline.
- Presented feature-importance findings, model tradeoffs, and forecasting implications to S&P Global stakeholders in senior-level reviews.